

Xue Wang

xwang297@jhu.edu | xwang297.github.io

Last update: October 1, 2026

EDUCATION

Johns Hopkins University

Baltimore, United States

PhD in Educational Evaluation and Research

Aug 2021 – May 2026

- Research interests: Learner autonomy and self-regulated learning; Literacy and language learning; AI and technology in education; Program evaluation; Meta-analysis
- Dissertation: *Measuring and Supporting Learner Autonomy, and Examining the Mediating Role of Engagement in the Relationship Between Preference for Autonomy and Reading Achievement*

The University of Hong Kong

Hong Kong SAR, China

MA in Teaching English to Speakers of Other Languages (Distinction)

Sep 2019

- Thesis: *Effectiveness of Learner Development in Tertiary English Education: A Meta-analysis*

Shantou University

Shantou, China

BA in English Language and Literature

Jul 2018

Dublin City University

Dublin, Ireland

Fully Funded Exchange Program in Applied Linguistics

Dec 2017 – Feb 2018

PUBLICATIONS

Peer-Reviewed Journal Publications (8)

1. Phan, L., Gatti, A., Han, Z., Li, N., ... **Wang, X.**, ... Hendrycks, D. (2026). A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649, 1139–1146. <https://doi.org/10.1038/s41586-025-09962-4> (IF: 64.8)
 - Contributing author among 1,000+ expert researchers; contributed education-domain questions to this AI benchmark dataset published in *Nature*
2. Xie, L., **Wang, X.**, & Jiang, L. (2026). Effects of GenAI-assisted digital multimodal composing on students' L2 writing performance: A quasi-experimental study. *TESOL Quarterly*. Advance Online Publication. (IF: 5.1)
3. **Wang, X.**, Zhang, Q. & Davis, M. H. (2025). Positive teacher feedback and adolescents' reading self-efficacy: A quasi-experimental analysis using PISA 2018. *Large-scale Assessments in Education*, 13. <https://doi.org/10.1186/s40536-025-00253-y> (IF: 3.0)
4. **Wang, X.**, Smith, G., & Watson, A. (2025). Self-regulated learning and self-efficacy among homeschooled students: A systematic review. *Journal of School Choice*. Advance Online Publication. <https://doi.org/10.1080/15582159.2025.2575559> (CiteScore: 2.1)
5. **Wang, X.**, Zhang, Q., Chen, H., Neitzel, A., & Davis, M. (2024). The effects of language learning strategy instruction on college students' English achievement and learner autonomy in mainland China: A meta-analysis. *International Journal of Educational Research*, 127. <https://doi.org/10.1016/j.ijer.2024.102442> (IF: 2.5)
6. **Wang, X.**, Neitzel, A., & Madden, N. (2024). Lightning Squad: Assessing the dosage effect of computer-assisted tutoring with cooperative learning for struggling readers. *Journal of Education for Students Placed at Risk*, 30(3), 254-272. <https://doi.org/10.1080/10824669.2024.2388242> (IF: 1.2)

7. **Wang, X.** (2022). Segmental versus suprasegmental: Which one is more important to teach? *RELC Journal*, 53(1), 194-202. <https://doi.org/10.1177/0033688220925926> (IF: 6.2)
8. Kung, F. W., & **Wang, X.** (2019). Exploring EFL learners' accent preferences for effective ELF communication. *RELC Journal*, 50(3), 394-407. <https://doi.org/10.1177/0033688218765306> (IF: 6.2)

Peer-Reviewed Conference Proceedings (2)

1. **Wang, X.**, & Luo, G. (2026). MetaMate: Understanding how educational researchers experience AI-assisted data extraction for systematic reviews. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. ACM. <https://doi.org/10.1145/3772363.3798755>
2. **Wang, X.** (2022). Hedging in academic writing: Cross-disciplinary comparisons in the Michigan Corpus of Upper-Level Student Papers (MICUSP). In T. D. Cooper & J. York (Eds.), *Japanese Association for Language Teaching Computer-Assisted Language Learning 2021 Conference Proceedings* (pp. 125-142). <https://doi.org/10.37546/JALTSIG.CALL.PCP2021-09>

Preprints (3)

1. **Wang, X.**, & Luo, G. (2026). Evaluating racial bias in LLM reasoning: Implications for equitable AI use in education. *EdArXiv*. https://doi.org/10.35542/osf.io/ynt3h_v1
2. Mehr, A., Howard, J., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., **Wang, X.**, Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). Improving evidence synthesis with artificial intelligence. *MetaArXiv*. https://doi.org/10.31222/osf.io/6d2wm_v1
3. **Wang, X.**, & Luo, G. (2024). MetaMate: Large Language Model to the rescue of automated data extraction for educational systematic reviews and meta-analyses. *EdArXiv*. <https://doi.org/10.35542/osf.io/wn3cd>
 - First AI tool designed for automated data extraction in education systematic reviews

Research Reports (1)

Neitzel, A. J., Storey, N., & **Wang, X.** (2026). *Air Reading: A randomized evaluation of a virtual tutoring model in Louisiana and Texas schools*. Center for Research and Reform in Education, The Johns Hopkins University. <https://jscholarship.library.jhu.edu/handle/1774.2/71611>

Public Scholarship (2)

1. **Wang, X.**, & Gehlbach, H. (August 2024). AI-created quizzes can save teachers time while boosting student achievement. *The 74*. <https://www.the74million.org/article/ai-created-quizzes-can-save-teachers-time-while-boosting-student-achievement/>
2. **Wang, X.**, & Fang, F. (December 2018). Awakening to Global Englishes. *EFL Magazine*. <https://eflmagazine.com/awakening-to-global-englishes/>

Under Review & In Progress (6)

1. Zhang, Q., **Wang, X.**, Tian, X., Gehlbach, H. Social support at your fingertips: A meta-analysis on the correlation between social media usage and social support. (Under review)
2. Xie, L., **Wang, X.**, Jiang, L., & Lai, C. The development and validation of a second language writing autonomy scale from a New Literacy Studies perspective. (Under review)
3. **Wang, X.**, Xie, L., Davis, M., Jiang, L. Scaffolding second language writing autonomy: A systematic review and meta-analysis of intervention effectiveness and design principles. (Under review)

4. Liu, J., Liu, L., Jiang, W., & **Wang, X.** Assessing the performance of Bayesian and frequentist estimators in confirmatory factor analysis with sparse Likert-scale data. (Revise and Resubmit)
5. Wang, F., **Wang, X.**, & Wang, J. Artificial intelligence for automated feedback in education: A meta-analysis. (Manuscript in preparation)
6. Liu, J., **Wang, X.**, & Wang, H. Pediatric Symptom Checklist-17: A systematic review of validation evidence. (Manuscript in preparation)

Doctoral Dissertation

Paper 1: A systematic review of instruments measuring learner autonomy in school-aged children and adolescents

Paper 2: A meta-analysis of interventions to support learner autonomy in K-12

Paper 3: Pathways to reading success: Examining the mediating role of engagement in the relationship between preference for autonomy and reading achievement

INVITED TALKS (1)

Wang, X. (2026, September 25). MetaMate: Automating study coding for educational systematic reviews with large language models [Invited online seminar]. American Educational Research Association Systematic Review and Meta-Analysis Special Interest Group Online Seminar Series.

SELECTED CONFERENCE PRESENTATIONS (21)

1. **Wang, X.**, Smith, G., & Watson, A. (2026, April 8–12). Self-regulated learning and self-efficacy among homeschooled students: A systematic review [Symposium session]. 2026 American Educational Research Association Annual Meeting, Los Angeles CA, United States.
2. **Wang, X.**, Hao, Y., Huang, Z., Bao, H., & Davis, M. (2026, April 8–12). Measuring learner autonomy in K-12: A systematic review of assessment instruments [Poster session]. 2026 American Educational Research Association Annual Meeting, Los Angeles CA, United States.
3. **Wang, X.**, & Luo, G. (2025, October 8–11). MetaMate 2.0: A customizable, open-source large language model tool for automated data extraction in educational systematic reviews [Paper session]. Society for Research on Educational Effectiveness 2025 Conference, Chicago IL, United States.
4. **Wang, X.**, & Luo, G. (2025, October 8–11). Evaluating racial bias in LLM reasoning: Implications for equitable AI use in education [Paper session]. Society for Research on Educational Effectiveness 2025 Conference, Chicago IL, United States.
5. **Wang, X.**, & Luo, G. (2025, April 23–27). MetaMate: Large Language Model to the rescue of automated data extraction for educational systematic reviews and meta-analyses [Paper session]. 2025 American Educational Research Association Annual Meeting, Denver CO, United States.
6. Xie, L., **Wang, X.**, Jiang, L., & Lai, C. (2025, April 23–27). The development and validation of a second language writing autonomy scale. [Roundtable session]. 2025 American Educational Research Association Annual Meeting, Denver CO, United States.
7. **Wang, X.**, & Luo, G. (2024, September 18–21). MetaMate: Large Language Model to the rescue of automated data extraction for educational systematic reviews and meta-analyses [Paper session]. Society for Research on Educational Effectiveness 2024 Conference, Baltimore MD, United States.
8. **Wang, X.**, Bao, H., Hao, Y., & Davis, M. (2024, April 11–14). A systematic review of learner autonomy scales: Mapping the landscape of measurement tools in autonomy interventions [Poster session]. 2024 American Educational Research Association Annual Meeting, Philadelphia PA, United States.

9. Zhang, Q., **Wang, X.**, Tian, X., & Gehlbach, H. (2024, April 11–14). Social support at your fingertips: Exploring the correlation between students' social media usage and social support through meta-analysis [Paper session]. 2024 American Educational Research Association Annual Meeting, Philadelphia PA, United States.
10. **Wang, X.** (2023, September 21–24). The effects of positive teacher feedback on student self-efficacy: A causal analysis [Paper session]. Society for Research on Educational Effectiveness 2023 Conference, Washington DC, United States.
11. Wang, J., **Wang, X.**, Zhang, Q., Bao, H., Zhao, Z., Neitzel, A. (2023, September 27–30). Risk of bias tools in systematic reviews of interventions PK-12 education: A critical analysis of studies in Review of Educational Research [Paper session]. Society for Research on Educational Effectiveness 2023 Conference, Washington DC, United States.
12. **Wang, X.** (2023, April 13–16). Critical pedagogy as a pathway to learner autonomy [Roundtable session]. 2023 American Educational Research Association Annual Meeting, Chicago IL, United States.
13. **Wang, X.**, & Wang, J. (2023, April 13–16). The effects of additional instruction on students' achievement in Greece and South Korea: Evidence from PISA 2018 [Paper session]. 2023 American Educational Research Association Annual Meeting, Chicago IL, United States.
14. **Wang, X.** (2023, April 13–16). The effects of private tutoring on Korean students' math achievement based on PISA 2018 data [Roundtable session]. 2023 American Educational Research Association Annual Meeting, Chicago IL, United States.
15. Nehring, L. D., Mu, N., **Wang, X.**, Zhang, Q., Cornwall, K., Gehlbach, H. (2023, April 13–16). Outcomes of an outdoor, residential learning experience in the context of pandemic and learning loss [Roundtable session]. 2023 American Educational Research Association Annual Meeting, Chicago IL, United States.
16. **Wang, X.**, Ochoa, A., Wang, J., Zhang, Q., Wang, F., & Neitzel, A. (2023, April 13–16). Effects of school-based autonomy-supportive interventions on academic outcomes and learning autonomy: A meta-analysis [Poster Session]. 2023 American Educational Research Association Annual Meeting, Chicago IL, United States.
17. **Wang, X.**, & Davis, M. (2023, Feb 18–22). The relationship between additional instruction and math achievement in South Korea: Evidence from PISA 2018 [Paper session]. Comparative and International Education Society 2023 Annual Conference, Washington, D. C., United States.
18. **Wang, X.** (2022, September 21–24). Inequality in the shadow? Cross-national comparisons of the effects of additional instruction on math and science performance based on PISA 2018 data [Paper session]. Society for Research on Educational Effectiveness 2022 Conference, Washington DC, United States.
19. **Wang, X.**, Neitzel, A., Ross, S. Laurenzano, M., Madden, N. (2022, September 21–24). Lightning Squad: Evaluating the efficacy of computer-assisted tutoring with cooperative learning for struggling readers [Poster session]. Society for Research on Educational Effectiveness 2022 Conference, Washington DC, United States.
20. **Wang, X.**, Chen, H., Zhang, Q., & Neitzel, A. (2022, August 4–6). Teach a man to fish: A meta-analysis on the effectiveness of strategy training on learning autonomy [Poster session]. American Psychological Association 2022 Annual Convention, Minneapolis, MN, United States.
21. **Wang, X.** (2021, June 4–6). Hedging in academic writing: Cross-disciplinary comparisons in the Michigan Corpus of Upper-Level Student Papers (MICUSP) [Paper session]. Japanese Association for Language Teaching Computer-Assisted Language Learning 2021 Conference, Japan.

SERVICE

Leadership & Committee Roles

Review Committee Chair, Category 3: Outstanding Dissertation, 2027 American Educational Research Association Division H Outstanding Publication Competition, 2026-2027

Graduate Student Representative-at-Large, American Educational Research Association Systematic Review and Meta-analysis Special Interest Group, 2025-2026

Co-chair, American Educational Research Association Annual Meeting Division H Exhibit Booth Committee, 2023-2026

Division Representative, School of Education, PhD Student Advisory Committee to the Vice Provost for Graduate and Professional Education, Johns Hopkins University, 2023-2024

Journal Reviewing

Ad hoc reviewer for 11 manuscripts (15 review reports, including revisions) across 8 peer-reviewed journals, 2020-2026:

[*Metacognition and Learning*](#), [*Review of Education*](#), [*Research Synthesis Methods*](#), [*International Journal of Educational Research*](#), [*The Asia-Pacific Education Researcher*](#), [*Journal of Psychoeducational Assessment*](#), [*Journal of School Choice*](#), [*Language Teaching Research*](#)

Conference Reviewing

Society for Research on Educational Effectiveness 2025 Conference, Chicago IL, United States (2 proposals reviewed).

American Educational Research Association Annual Meetings, 2023-2027 (Studying and Self-Regulated Learning Special Interest Group; 21 proposals reviewed)

American Educational Research Association Annual Meetings, 2027 (Systematic Review and Meta-Analysis Special Interest Group; 11 proposals reviewed)

American Educational Research Association Annual Meeting, 2023 (Division C-Learning and Instruction; 4 proposals reviewed)

Science Communication & Outreach

Contributor, [*Best Evidence in Brief Live*](#) (webcast), The National College and Center for Research and Reform in Education at Johns Hopkins University, 2022-2023

Writer, [*Best Evidence in Brief*](#) (bi-weekly e-newsletter), 2021-present

GRANTS & HONORS

Fellowships

Johns Hopkins University School of Education Quantitative Research Fellowship, 2021-2025

- Awarded \$138,800 to support doctorate study, research, and professional development.

New York University Steinhardt Faculty First-Look Fellow, 2025, (\$1,000)

Awards

Winner, AERA Division H Outstanding Publication Competition Award, Outstanding Dissertation Category, 2026, (\$1,000)

Johns Hopkins School of Education Excellence Award in the PhD Program, 2026

Research Grants

OpenAI Researcher Access Program, 2024, (\$5,000 in API credits)

- Supported the development and validation of [*MetaMate*](#), an AI application designed to assist with automated data extraction in educational research syntheses.

AERA Division H Graduate Student Research Grant, 2022-2023, (\$1,500)

- Supported data collection for a meta-analysis on autonomy-supportive interventions in K-12.

Grant Proposals Under Review

National Science Foundation, Collaborative Research Proposal, 2026 (Submitted; Role: Methodologist)

- Project MAP (Misconceptions, Assessment, and Practices)-Fractions: An Investigation of Students' Common Misconceptions in Learning Fractions and Instructional Practices to Address Them. PI: Dr. Jihyun Lee (University of Wyoming); Co-PI: Dr. Lina Shanley (University of Oregon).

Institute of Education Sciences, 2026 (Submitted; Role: Key Personnel)

- MetaMate 2.0: Reliable AI-Assisted Study Coding and Data Extraction for Educational Systematic Reviews and Meta-Analyses. Proposal to further develop MetaMate, the AI tool I created for automated data extraction in educational research syntheses. PI: Dr. Amanda Neitzel (Johns Hopkins University); Co-PI: Dr. Ju Sun (University of Minnesota, Twin Cities).

Travel Awards

American Educational Research Association, Studying and Self-Regulated Learning Special Interest Group Student Travel Award, 2026, (\$500)

American Educational Research Association, Division H Student Travel Award, 2026, (\$130)

Society for Research on Educational Effectiveness Conference Attendance Grant, 2025, (\$700)

TEACHING & MENTORING

Instructor of Record

Advanced Instructional Strategies | Adjunct Faculty

Baltimore, USA

MS in International Teaching and Global Leadership, Johns Hopkins

Jan 2024 – May 2024

- Instructed 20 students on identifying, explaining, developing, and applying effective instructional strategies tailored to diverse learner populations.

Teaching Assistantships

Intermediate Statistics

Baltimore, USA

MS in Education Policy, Johns Hopkins University

Jan 2025 – May 2025

- Facilitated weekly office hours to address student questions on data handling and interpretation.

Data Workflow

Baltimore, USA

PhD in Education, Johns Hopkins University

Jan 2024 – May 2024

- Designed and delivered a methodology workshop on missing data analysis, covering missing data mechanisms and imputation techniques.
- Led weekly office hours to provide hands-on support for students' R coding projects.

Evaluation of Education Policies and Programs

Baltimore, USA

Doctor of Education, Johns Hopkins University

Jan 2024 – May 2024

- Designed and delivered a methodology workshop on power analysis and effect size calculation.
- Addressed student inquiries on research design and statistical analyses.

Diversity in American Education

Baltimore, USA

MS in Education Policy, Johns Hopkins University

Aug 2023 – Dec 2023

- Designed and delivered two academic writing workshops.
- Facilitated online student group discussions.

Mentoring

Extended Learning | Teacher Mentor

Baltimore, USA

MS in International Teaching and Global Leadership, Johns Hopkins

Oct 2022 – May 2024

- Led weekly mentoring sessions for 13 graduate students on conducting meta-analyses.
- Conducted bi-weekly workshops on introductory statistical analysis using R.

WORK EXPERIENCE

Research Program Analyst

Baltimore, USA

Center for Research and Reform in Education, Johns Hopkins University

Jul 2026 – Present

- Evaluating early reading and math interventions for students performing below grade level, primarily through randomized controlled trials as well as quasi-experimental and mixed-methods designs, using hierarchical linear modeling and propensity score matching.

R Programming Analyst

Baltimore, USA

Vision for Baltimore Project

Nov 2022 – Present

- Developing R scripts to manage, analyze, and report data for stakeholders in the Vision for Baltimore project, a school-based vision program offering vision services to pre-K to 8th-grade students in Baltimore City public schools.

Research Assistant

Hong Kong, China

The Education University of Hong Kong, Department of Education Policy

Oct 2019 – Aug 2021

- Conducted qualitative research on educational leadership and supported school improvement programs.

FUNDED PROFESSIONAL DEVELOPMENT

2024 National Assessment of Educational Progress (NAEP) Training

Arlington, USA

- Selected to attend an intensive three-day workshop funded by the National Center for Education Statistics (NCES) on data analysis strategies using large-scale assessment data such as NAEP data (funding: \$1,500).

2024 NSF-funded Network for Invivo Research (NIER) Workshop

Philadelphia, USA

- Selected to attend one half-day workshop to contribute to the NIER initiative to build infrastructure supporting teacher-student research (funding: \$300).

2023 Meta-analysis Training Institute (MATI) Workshop

Chicago, USA

- Selected to attend an intensive one-week workshop funded by the Institute of Education Sciences (IES) on advanced methods for conducting large-scale research synthesis and meta-analysis (funding: \$2,000).